CHUKWUMA Clifford NWANNA

Electronics Engineer | Data Scientist | AI/ML Engineer

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PROFESSIONAL SUMMARY

Electronics and Computer Engineer with 3+ years of progressive experience spanning medical electronic systems installation and field service across 20+ Nigerian states, enterprise AI and data science at a top-tier financial institution, and award-winning embedded systems design. At Wema Bank, I design and maintain production-grade AI systems on Microsoft Azure, including a churn prediction pipeline scanning 320,000+ corporate accounts monthly and a governed natural-language-to-SQL platform used by business teams across the bank. My engineering foundation is built through biomedical field operations, hardware design projects recognized at national engineering competitions, and federal government research institution training. I have a practical, systems-level approach to designing reliable, regulated, and scalable technology solutions across both hardware and intelligent software domains.

KEY SKILLS

- Electronic systems design, installation & commissioning
- Medical equipment fault diagnosis & repair
- Machine learning & predictive modelling
- LLM / Generative AI application development
- Azure cloud platforms (ML, OpenAI, Synapse)
- Requirements analysis & stakeholder engagement
- ETL pipelines & data engineering (Python, SQL)
- Embedded systems (Arduino, ESP32, C++)
- Solar PV systems & power electronics
- Technical documentation & end-user training
- Cross-functional team leadership

PROFESSIONAL EXPERIENCE

Feb 2025 – Present | **Wema Bank Plc**, Lagos, Nigeria
Data Scientist | AI/ML Engineer

Key Responsibilities

- Design and deploymet of enterprise AI systems including churn prediction pipelines, natural-language-to-SQL engines, and LangChain-powered intelligence agents on Microsoft Azure (Azure ML, Synapse Analytics, Azure OpenAI, Azure App Service).
- Gathering of business requirements from IT, Compliance, and product stakeholders; translating them into governed technical specifications for regulated AI systems subject to CBN data standards.
- Lead end-to-end analytics pipelines covering data extraction, feature engineering, model development, deployment, and monitoring using Python, T-SQL, FastAPI, and Azure services.
- Delivering of cross-functional Agile sprint cycles, presentation of system outcomes to bank leadership, and management of stakeholder expectations through structured reporting.
- Design and implementation of security architectures including Row-Level Security injection and Semantic Firewall patterns to ensure AI outputs remain within approved data governance boundaries.

Key Achievements

- Deployed CRIS (Corporate Risk Intelligence System), a Python/SQL churn prediction pipeline on Azure ML + Synapse Analytics that scans 320,000+ corporate accounts and identifies 5,000–6,000 monthly intervention targets for the bank's Relationship Management teams.
- Built Data Knight, a governed NL-to-SQL engine using Azure OpenAI and FastAPI that reduced ad hoc data request turnaround from days to minutes for non-technical business users.
- Designed the Semantic Firewall architecture. A reusable security primitive now deployed across multiple Wema Bank AI products ensuring AI agents cannot generate queries outside pre-approved boundaries.

Sep 2022 – Feb 2025 | **Coscharis Medical and Foods Limited**, Lagos, Nigeria
Field Service Engineer

Key Responsibilities

- Installed, commissioned, and maintained a wide range of medical electronic equipment including ultrasound systems (SonoScape S40, Esaote MyLab30 GOLD), flat panel detectors (Tornel T-1717w), electrocardiographs (GE Venue Go R4w, Carl Novel AK3), ventilators, patient monitors, and pharmaceutical refrigeration units across hospitals, clinics, and university research laboratories in 20+ states.
- Performed systematic multi-level fault diagnosis on complex medical systems: isolating hardware faults at component level (motherboard replacement, board-level isolation testing), identifying software license failures and configuration errors, and restoring full clinical functionality.
- Installed and configured flat panel detector software (ZManager + YLDR), including network configuration, hardware calibration at 70kV/6.3mAs exposure settings, and training of clinical radiographers and technicians.
- Provided virtual troubleshooting support via remote video call for geographically distant facilities, resolving hardware faults without requiring field travel — including BIOS-level hard drive diagnosis and kernel recovery for SSI-5000 ultrasound scanners.
- Designed and delivered a 16-week structured intern training curriculum covering biomedical engineering principles, embedded systems, IoT, data acquisition, and field operations; supervised and evaluated junior technical interns on-site.
- Produced signed field service reports, installation certificates, and technical documentation for all engagements, maintaining a complete audit trail across 50+ documented service visits.

Key Achievements

- Diagnosed and resolved four concurrent faults on an Esaote MyLab30 echocardiogram (trackball failure, board incompatibility, image board fault, and expired software license) restoring a system that had been non-functional at a cardiology center.
- Commissioned a Haier Thermocol HYC-360 pharmaceutical refrigerator at Ahmadu Bello University molecular laboratory, diagnosing and resolving an E5 control unit error via service manual cross-reference after an 8-hour site stabilization period.
- Successfully replaced a faulty control board in an Esaote MyLab30 echocardiogram under remote guidance from an OEM specialist (Engr. Toby Graff), restoring full clinical functionality to a cardiac diagnostic facility.

Jul 2019 – Dec 2019 **Electronics Development Institute (ELDI — NASENI)**, Awka, Anambra State, Nigeria
Electronics Intern (Federal Government Research & Development Institution)

Key Responsibilities

- Completed structured daily technical training across analogue circuit design, digital electronics, Arduino embedded programming, solar energy systems, and electronic troubleshooting and repair.
- Designed and constructed assigned circuits and systems on breadboard, applying component calculations, testing, and documentation to each completed assignment.
- Participated in field installation of a 3.5 KVA solar inverter system from panel mounting, DC wiring, charge controller connection, battery bank sizing, and load segregation analysis under supervision of Engr. Uche Ahjuzie.
- Conducted independent research on Light Dependent Resistor characteristics, evaluating LDR modules as design alternatives and presenting findings to the supervising engineer.

Key Achievements

- Independently designed and constructed a fully functional automatic nightlight system (LDR + LM358 op-amp comparator + D965 transistor + 12V relay + 7805 dual-rail power supply), identifying and resolving component tolerance discrepancies through systematic iterative calibration using variable resistors.
- Achieved a grade of A in Industrial Training defense assessment, formally evaluated by the Nnamdi Azikiwe University engineering panel.

EDUCATION

Bachelor of Engineering (B.Eng.) — Electronics & Computer Engineering

GOMYCODE Data Science Bootcamp

Nnamdi Azikiwe University, Awka, Nigeria
Graduated May 2021

GOMYCODE (State-approved Training Centre No. 11-1940-2)
Certificate of Completion | Aug 2023 – Feb 2024

TRAINING & OTHER SKILLS

- Wisonic Medical Ultrasound Product Training — Clover, Navi, Piloter product lines (May 2023)
- UPS Systems Knowledge Training — Shenzhen Upsen Electronic Co., Ltd. (Apr 2024)
- Microsoft Azure AI Fundamentals (AI-900)
- Microsoft Azure Fundamentals (AZ-900)
- GitHub Foundations Certification
- Google Advanced Data Analytics Certificate
- IBM Data Science Professional Certificate
- Proficient: Python, SQL (T-SQL), C++, JavaScript/TypeScript, LangChain, Azure OpenAI, FastAPI, Scikit-learn, RAG pipelines, pgvector, Arduino/ESP32 embedded C++
- Proficient written and spoken English

PROFESSIONAL MEMBERSHIPS

IEEE – Professional Membership
NSE – Nigerian Society of Engineers, Graduate Membership

AWARDS

1st Place (Regional) + 2nd Place (National) — CODET 5th National Engineering Students Competition, 2021
3rd Place — Undergraduates Projects Challenge — SPE Annual STEM/AI Fair, Port Harcourt, May 2021

References available on request.